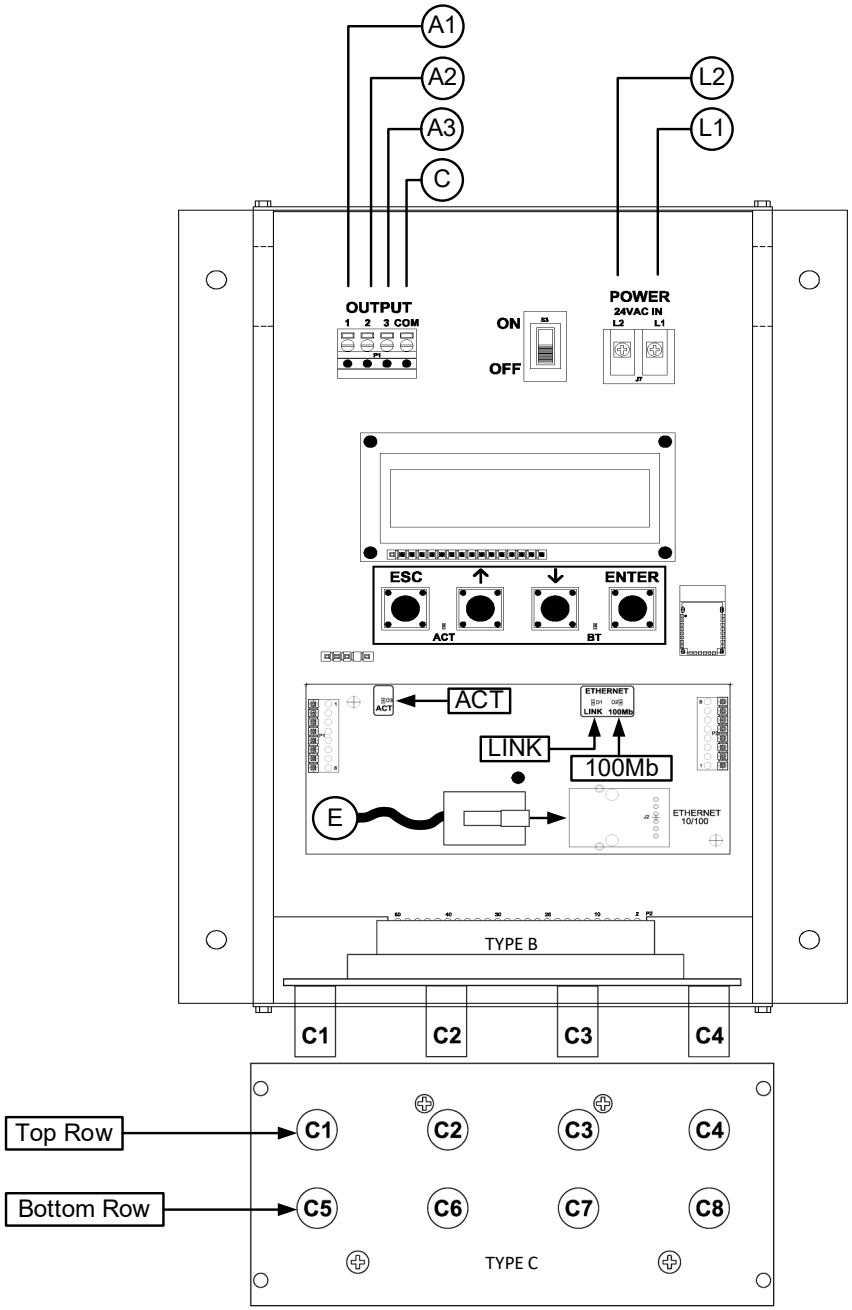
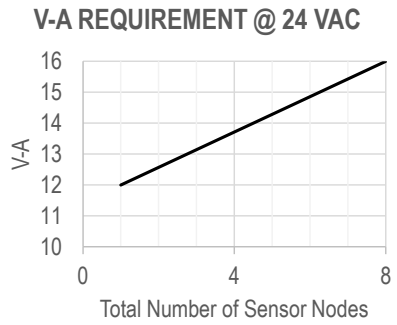
Advantage IV (A4) GTM108e-F WIRING GUIDE



TRANSMITTER CONNECTIONS

Power		Analog Out (isolated)				Ethernet	Type	
L1	L2	A1	A2	A3	C	E	B	C
24 VAC (hot)	24 VAC (neutral)	Airflow +	Temperature or Alarm +	N/A	Signal Common	RJ-45 CAT5 or greater	4 probes x 1 sensor/probe	8 probes x 1 sensor/probe



INSTRUCTIONS TO INSTALLER:

1. Mount the transmitter in a location where all probe cables can reach the receptacles of the transmitter. Provide a weatherproof enclosure (by others) and mount away from direct sunlight when outdoor mounting is required.
2. Connect the sensor probes to the transmitter. Although probes are "plug and play" and connections to specific receptacles are not required, it is recommended that probes are connected Probe 1 to receptacle C1, Probe 2 to receptacle C2, etc. Probe numbers are indicated on each cable hang tag.

⚠ *Fan array models (-F/An) must follow this convention if two probes are located in each fan inlet and the individual fan airflow rates are required and/or the fan alarm is enabled (i.e. put probes for fan 1 in C1 and C2, fan 2 in C3 and C4, etc.).*

ⓘ *Cables have an FEP plenum rated jacket that are UV tolerant and suitable for operation over the entire operating temperature range of the device.*

⚠ *Sensor probe plugs are keyed and NOT twist-lock. Align the key and push the plug onto the transmitter receptacle. Twisting may damage the connector pins.*

3. Select a 24 VAC transformer that provides 22.8 to 26.4 VAC during operation. Refer to the chart above to optimize the transformer size or size the transformer for 16 V-A for each measurement location.

⚠ *Multiple transmitters wired to a single transformer must be wired "in-phase" (L1 to L1 and L2 to L2).*

4. If analog output signals are used, continue to step 5, otherwise skip to step 6.
5. Connect each analog output signal required to the host B.A.S. using shielded twisted-pair wire. Properly terminate the shield (typically at the B.A.S.).

⚠ *If twisted pair wire and/or shielded cable is not used, extraneous electrical noise can be picked up between the transmitter and host control panel.*

6. If the Ethernet connection is required continue to step 7, otherwise skip to step 8.
7. Connect to an Ethernet network (BACnet Ethernet, BACnet IP, Modbus TCP, TCP/IP or *EB-Bus*), if required, using a standard RJ-45 terminated Ethernet cable, CAT5 or greater.

⚠ *Use of improperly wired RJ-45 connectors may damage the Ethernet output circuit of the transmitter.*

8. Refer to the *GTM108e-F Startup Guide* prior to moving the power switch to the "ON" position.